Model vs Index — CAGR Comparison

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What period this covers

Item	Value
Test window	2025-02-01 to 2025-10-31
Rebalance frequency	monthly
Rebalance dates	8
Elapsed time	0.67 years
Holding period	entry to the next rebalance's entry (chaining)
Portfolio	equal-weighted top 20, rebalanced each period
CAGR formula	product of (1 + period return) over 8 periods, then raised to 1/0.67

Basis

Price return compared with price return. NSE's historical index endpoint serves price indices only — every total-return (TRI) spelling returns no data. The strategy's dividends are therefore reported separately rather than folded into the headline, because comparing a dividend-inclusive strategy against a dividend-less index would credit the model with roughly the market's yield (about 1–1.5% a year in India) for nothing.

Index returns are measured over the sessions the strategy actually traded — the same next-session entry and the same exit — not between month-ends, so neither side is measuring a different span of market time.

Headline: CAGR by strategy

Strategy	Gross	Net	vs UNIVERSE	vs N50	vs N500	vs NMIDCAP 150	vs NSMALLCAP 250
equal weight universe	34.42%	32.05%	-2.37%	+6.35%	+1.14%	-7.64%	-10.64%
growth only	58.87%	56.45%	+22.03%	+30.75%	+25.54%	+16.76%	+13.76%
model 5	39.80%	33.16%	-1.26%	+7.45%	+2.25%	-6.53%	-9.53%
model 5 gated	22.99%	16.89%	-17.53%	-8.82%	-14.02%	-22.80%	-25.80%
momentum only	19.78%	10.74%	-23.68%	-14.97%	-20.17%	-28.95%	-31.95%
quality only	18.30%	17.18%	-17.24%	-8.52%	-13.73%	-22.51%	-25.51%
random ranking	45.49%	21.23%	-13.19%	-4.47%	-9.68%	-18.46%	-21.46%
simple quality momentum	36.46%	31.01%	-3.40%	+5.31%	+0.10%	-8.67%	-11.68%
value only	61.05%	55.85%	+21.43%	+30.14%	+24.94%	+16.16%	+13.16%

vs UNIVERSE is the column that matters, and it is deliberately placed before the index columns. It compares the strategy against an equal-weighted holding of the very same point-in-time universe it selected from, so it isolates stock selection.

Why the index columns flatter every strategy. The eligible universe returned 34.42% against NIFTY 500's 30.91% — a +3.51 point gap earned purely by equal-weighting a broad, small-cap-heavy universe, before any selection at all. That gap is inside every "vs index" number below. A strategy that beats NIFTY 500 by 20 points while matching the universe has selected nothing.

Index reference

Index	CAGR	Volatility	Sharpe	Max drawdown	Periods
NIFTY 50	25.70%	10.17%	2.31	-3.82%	8
NIFTY 500	30.91%	10.89%	2.55	-3.99%	8
NIFTY MIDCAP 150	39.69%	13.12%	2.64	-3.93%	8

NIFTY SMALLCAP 250	42.69%	17.66%	2.12	-5.76%	8
Eligible universe (equal weight)	34.42%	-	-	-	8

The **eligible universe** row is the benchmark that controls for size and breadth: the same point-in-time universe the strategy chose from, held equally weighted. Beating NIFTY 500 while trailing this means the model captured a small-cap tilt rather than stock selection.

Risk-adjusted comparison

Strategy	Basis	CAGR	Volatility	Sharpe	Max DD	Hit rate
equal weight universe	net	32.05%	14.42%	2.01	-5.11%	75%
growth only	net	56.45%	19.89%	2.38	-4.99%	75%
model 5	net	33.16%	16.02%	1.88	-7.36%	75%
model 5 gated	net	16.89%	17.94%	0.95	-7.54%	50%
momentum only	net	10.74%	15.97%	0.71	-7.99%	50%
quality only	net	17.18%	10.22%	1.61	-3.78%	62%
random ranking	net	21.23%	24.90%	0.88	-11.32%	62%
simple quality momentum	net	31.01%	17.43%	1.64	-3.85%	50%
value only	net	55.85%	13.60%	3.38	-1.22%	88%

Excess return against NIFTY 500

Strategy	Mean excess/period	Tracking error	Information ratio	Periods beaten
equal weight universe	+0.10%	11.25%	0.11	50%
growth only	+1.62%	15.74%	1.24	62%
model 5	+0.19%	7.37%	0.32	38%
model 5 gated	-0.89%	13.99%	-0.76	38%
momentum only	-1.37%	13.38%	-1.23	38%
quality only	-0.95%	7.88%	-1.44	50%
random ranking	-0.48%	19.91%	-0.29	38%
simple quality momentum	+0.07%	14.25%	0.06	50%
value only	+1.52%	13.96%	1.30	62%

Information ratio is excess return per unit of tracking error — how reliably the outperformance was earned, not how large it was. **Periods beaten** is the share of rebalances where the strategy outpaced the index; a high CAGR with a low share was carried by a handful of periods.

Turnover and cost drag

Strategy	One-way turnover	Gross CAGR	Net CAGR	Cost drag
equal weight universe	0.089	34.42%	32.05%	2.37%
growth only	0.078	58.87%	56.45%	2.42%
model 5	0.361	39.80%	33.16%	6.65%
model 5 gated	0.456	22.99%	16.89%	6.10%
momentum only	0.506	19.78%	10.74%	9.04%
quality only	0.100	18.30%	17.18%	1.12%
random ranking	0.939	45.49%	21.23%	24.26%
simple quality momentum	0.356	36.46%	31.01%	5.44%
value only	0.139	61.05%	55.85%	5.20%

Cost drag is annualised, and it is the difference between a ranking that works on paper and one that works in an account. A strategy replacing most of its book every month pays it repeatedly.

Before quoting these numbers

An index is investable and this portfolio is a simulation. The index CAGR is achievable through a low-cost fund; the strategy CAGR assumes every fill happened at the modelled price and size.

The window is what it is. A CAGR over a few years is dominated by the market regime inside it. Check the periods-beaten share and the drawdown before treating a CAGR gap as skill.

Costs are modelled, not measured. Statutory charges are exact and effective-dated; half-spread and market impact are assumptions, which is why gross and net are both shown.

Delisting recovery is an assumption. Re-run with --delisting-strategy zero and last_close to see how much of the gap depends on it.